

Project Design Phase

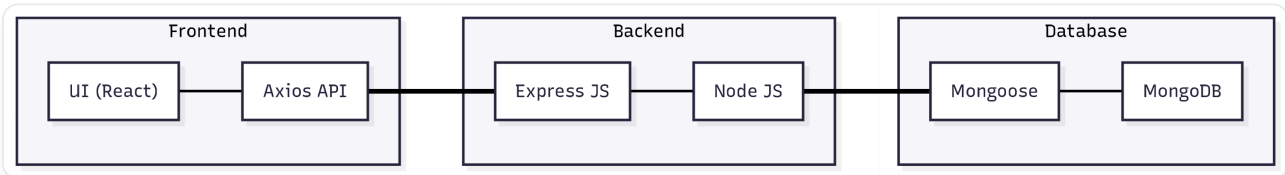
Solution Architecture

Date	18th June 2026
Team ID	
Project Name	MediCareBook (Book a Doctor)
Maximum Marks	4 Marks

Solution Architecture Overview & Guidelines:

The solution architecture provides a high-level representation of logical processing layers. Creating an effective structure requires mapping external API interfaces, client-to-server networks, database persistence tiers, and local/cloud hosting demarcations.

Project Implementation: MediCareBook Architecture



System Tier	Architectural Implementations
Presentation Tier	React client application styled with Tailwind CSS utility grids. Uses Axios fetch engines and caches user credentials in browser local storage. Routes matching user roles to mount dashboards.
Application Tier	Stateless Node.js and Express backend API servers. Protects endpoint routes using custom Token verification and RBAC middleware validators. Passes verified payloads to auth or doctor controller scripts.
Database Tier	MongoDB NoSQL document store mapped via Mongoose schemas. Enforces validation constraints on fields and utilizes pre-remove hook cascades to purge orphan booking documents upon user account deletions.